

# Tejas Vaity

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**Software Developer** with 3.5 years of experience in Java (backend), Angular (frontend) and Python(Flask/Django) building scalable web applications with a focus on robust, intuitive and high performance solutions.

## Education

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**University of Michigan**, MS in Computer and Information Science (GPA-3.88) Sep 2024 to May 2026

## Experience

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**Research Assistant**, University of Michigan - Michigan, USA May 2025

- Optimized the CPS system using Docker by introducing new techniques and rules to enhance performance and maintainability.
- Developing a Flask-based internal dashboard for automating CI/CD testing pipelines integrating GitHub Actions and Jenkins, improving research workflow efficiency by 30%.
- Built data analysis utilities using Python and Pandas, enabling automated research reports and visualization.

**Full Stack Developer**, NSEIT – Mumbai, India Feb 2022 to Jul 2024

- Architected and developed a high-performance, real-time trading platform using Java, Spring Boot, Angular, and Apache Kafka.
- Designed and implemented a scalable, responsive front-end web application using HTML5, CSS3, JavaScript, and TypeScript. Enhanced user experience and increased system uptime by 10%.
- Optimized real-time data processing and streamlined system integrations leveraging Microservices architecture, Design Patterns, Object-Oriented Programming(OOP) principles, RESTful APIs, and Socket Programming.
- Deployed on Linux servers with Tomcat, implemented Jenkins CI/CD pipelines, and improved system latency.
- Improved data retrieval speed by 25% by implementing Redis caching and optimizing database queries.
- Led and mentored a 6-member agile development team. Managed the full software development lifecycle(SDLC) from planning and execution to delivery.

**Software Frontend Developer**, Faclon Labs – Mumbai, India Oct 2021 to Feb 2022

- Enhanced UI with Angular Material, improving tables and graphs for intuitive data visualization.
- Resolved 95% of system issues within 24 hours, boosting system reliability and client satisfaction while refining web services performance.
- Boosted code efficiency by 20% using design patterns and modern development tools.

**Project Engineer**, Wipro Limited – Mumbai, India Mar 2021 to Sep 2021

- Analyzed system requirements to create test cases and documented defects found during the software development process, focusing on quality assurance for a banking application.
- Provided team support alongside testing to maximize software reliability and performance.

## Technologies

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**Programming Languages:** Java, C, Python, JavaScript, TypeScript, HTML, CSS, Python, Shell Scripting, Node.js

**Frameworks:** Angular, React, Spring Boot, Spring Security, Spring JPA, Spring MVC, Spring Cloud, Bash, Nest.js

**Databases:** SQL Server, Oracle, MySQL, Postgres, Redis, MongoDB

**Tools:** Apache Kafka, Tomcat, Jenkins, AWS, Android Studio, Docker, Jenkins, Jupyter Notebook

**Operating System:** Linux/Unix, Windows | **Version Control:** SVN, Git | **IDE:** Eclipse, Visual Studio code

## Projects

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- **Research Automation Tool** - Developed a Flask and Python web application to automate repository analysis and CI/CD testing workflows. Integrated with GitHub Actions and Jenkins, improving testing efficiency by 40%. Used SQLAlchemy for database management and Bootstrap for responsive UI.
  - **Penny Buddy** - Build user-facing dashboards for real-time financial data visualization using Nest.js and Plaid API, implementing responsive design principles and design patterns.
  - **BrickWise** - Developed a real estate web application using the MERN stack, enabling users to list properties for rent or sale, integrated with interactive maps and a real-time chat system.
  - **Retinal Disease Classification** - Developed and optimized deep learning models using Convolutional Neural Networks(CNNs) and TensorFlow/Keras for binary and multi-class retinal disease classification. On the RFMiD dataset implemented advanced techniques, including transfer learning, data augmentation, and image preprocessing.